

Hongchen (Steven) Yang

Waltham, MA | stevenyang0316@gmail.com | (704) 345-2919 | [LinkedIn](#) | [GitHub](#) | [Website](#)

Education

Brandeis University, Waltham, MA

Expected Dec 2026

B.S. in Computer Science GPA: 3.748 / 4.0

Dean's List (All Semesters); Provost's Summer Undergraduate Research Fellowship (Summer 2025)

Relevant Coursework: Computer Systems Design; Database Management; Operating Systems; Machine Learning; Theory of Computation

Research Experience

Smart & Scalable Data Systems (SSD) Lab, Researcher — Brandeis University

May 2025 – Present

- Study how workload shifts, skew, and burstiness affect RocksDB compaction and performance, combining recent LSM-tree literature with source-code analysis and benchmarking.
- Categorize tunable knobs by objectives such as throughput, latency, CPU utilization, and amplification, then prune dependent parameters to reduce configuration search cost.
- Investigate workload-aware knob selection using Bayesian optimization, tree-based surrogate models, and lightweight online learning for changing workloads.
- Build prototype benchmarking workflows to evaluate tuning policies and inform the design of *FluidLSM*, a self-adaptive LSM-based key-value store.

Data Science Intern (Independent Study) — Brandeis University

Jan 2026 – Present

- Design a controlled benchmark to measure how schema heterogeneity, missing data, and fragmented records degrade text-to-SQL performance on higher-education administrative data.
- Build shared synthetic schemas and controllable fragmentation generators, along with Python ETL and evaluation pipelines for reproducible comparisons between baseline and fragmented data. Analyze provenance and join/query quality.

Teaching Experience

Teaching Assistant, Introduction to Database — Brandeis University

Jan 2026 – Present

- Lead office hours and review sessions for about 50 students covering SQL, ER modeling, normalization, indexing, and query optimization.
- Grade assignments and exams and give feedback on query design, relational modeling, query optimization, and database system fundamentals.

Internship Experience

Hefei City Cloud Data Center Co., Ltd. — Hefei, China

AI Development Intern

Jun 2026 – Aug 2026

- Designed and implemented an end-to-end AI workflow that ingests reimbursement emails and attachments, recognizes documents, extracts structured fields, and generates data for downstream processing.
- Developed custom agent memory and vector-retrieval nodes that write, append to, clear, and persist memory and perform scoped semantic search across multi-turn tasks.
- Added memory-resource lifecycle APIs and UI to an agent workflow platform and integrated terminology-retrieval and report-writing nodes. Migrated reimbursement workflows and validated database, vector-retrieval, and model-service integrations.

Shanghai Development Center of Computer Software Technology — Shanghai, China

Software Engineering Intern

May 2024 – Aug 2024

- Traced Supersonic AI's text-to-SQL workflow from user prompts to generated queries across translation, parsing, planning, and execution stages.
- Analyzed the RuoYi backend architecture and its Maven-based integration of Spring Boot, MyBatis, and Redis.
- Used Linux development tools to run services, inspect logs, and debug backend tasks.
- Applied Java streams and lambda expressions to refactor service logic and improve code readability and maintainability.
- Analyzed how data structures and algorithmic choices affect backend performance.

Projects

Blacklight Privacy Detection Tool — The Markup & Brandeis University

Aug 2025 – Dec 2025

[Blacklight Update Blog](#)

- Contributed to the development and validation of Blacklight updates that detect TikTok, X, and other third-party tracking pixels.
- Collaborated with Brandeis and The Markup on Blacklight, a public privacy inspection tool that has handled more than 18 million scans.

Technical Skills

Languages: Java, C++, Python, Go, Rust, JavaScript, TypeScript, SQL

Frameworks & Systems: Spring Boot, MyBatis, RocksDB, PostgreSQL, MySQL, Redis, Docker, Linux, Git

Languages Spoken: Mandarin Chinese (Native), English (Fluent)